

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient is diagnosed with community-acquired pneumonia and is treated with cephalexin, which is an antibiotic. This suggests the infection is caused by a bacterial pathogen, most likely Streptococcus pneumoniae or another Streptococcus species. Streptococcus species are Gram-positive bacteria that have peptidoglycan as a major component of their cell wall. The major antigen that can induce an immune response in this context is peptidoglycan. This immune response is typically Th1-cell mediated and involves TLR4, not TLR5, which is more commonly associated with peptidoglycan from Gram-negative bacteria.

</think>

<answer> C. Peptidoglycan is its major antigen that induces an immune response. </answer>

